

Lecture 12: Higher Order Derivatives

GOAL: Compute derivatives of derivatives. Interpret the second and third derivative in a physics context.

Summary:

$$\frac{d^n f}{dx^n} = \frac{d}{dx} \left(\frac{d^{n-1} f}{dx^{n-1}} \right) \quad f^{(n)}(x) = [f^{(n-1)}(x)]'$$

$$a(t) = v'(t) = s''(t) \quad j(t) = a'(t) = v''(t) = s'''(t)$$

Recall that the height of a ball tossed straight into the air can be modeled using a parabola:

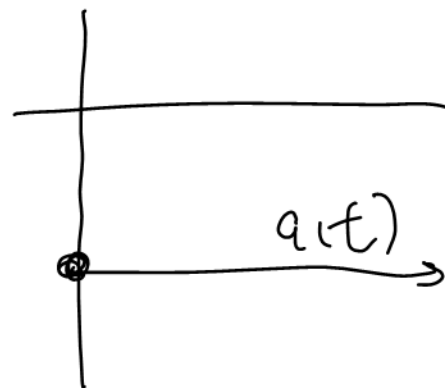
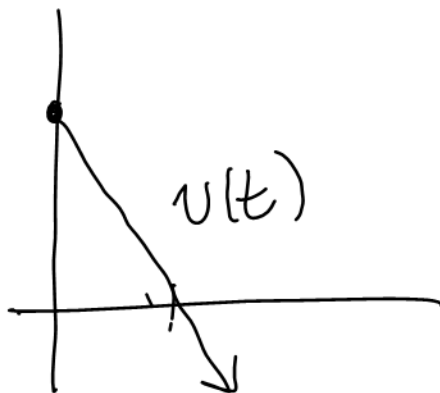
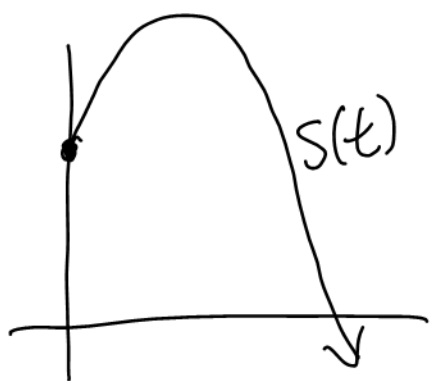
$$s(t) = -4.9t^2 + 7t + 6$$

It's velocity:

$$v(t) = -9.8t + 7$$

We can continue:

$$v'(t) = -9.8 \stackrel{\text{def}}{=} a(t)$$



Def The

quantity

$$\frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx^2}$$

$$[f'(x)]' = f''(x)$$

is called the 2^{nd} Derivative of f w.r.t. x

Example 1 Find $f''(x)$ if $f(x) = 2\sin x + 3\cos x$

$$f(x) = 2\sin x + 3\cos x$$

$$f'(x) = 2\cos x - 3\sin x$$

$$f''(x) = [f'(x)]' = -2\sin x - 3\cos x$$

$$= -(2\sin x + 3\cos x) = -f(x)$$

Def The acceleration of an object, whose position at time t is given by

$s(t)$, is defined by

$$a(t) \stackrel{\text{def}}{=} v'(t) = [s'(t)]' = s''(t)$$

Example 2

A position is traveling in a straight line; its position at time t is given by

$$s(t) = t^3 - 6t^2 + 9t$$

a) Find the velocity and acceleration of the particle at time t .

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

$$a(t) \stackrel{\text{def}}{=} v'(t) = 6t - 12$$

b) If t is measured in seconds and $s(t)$ is measured in meters, find $v(4)$ and $a(4)$.

$$v(4) = 3(4)^2 - 12(4) + 9 = 9 \frac{\text{m}}{\text{s}}$$

$$a(4) = 6(4) - 12 = 12 \frac{\text{m/s}}{\text{s}} = 12 \frac{\text{m}}{\text{s}^2}$$

c) What is the acceleration when the velocity is $24 \frac{\text{m}}{\text{s}}$?

Step 1 Find the time when $v(t) = 24$

$$v(t) = 3t^2 - 12t + 9 \quad \underline{\underline{\text{set } 24}}$$

$$3t^2 - 12t - 15 = 0$$

$$t^2 - 4t - 5 = 0$$

$$(t+1)(t-5) = 0$$

$\Rightarrow t = -1, 5$. As $t \geq 0$, $v(5) = 24$

Step 2 Find $a(t)$ when $t = 5$

$$a(t) = 6t - 12$$

$$a(5) = 6(5) - 12 = 30 - 12 = 18 \frac{\text{m}}{\text{s}^2}$$

Example 3

Suppose the position of a particle is given by

$$s(t) = \frac{t}{t^2 + 1} \quad (t \geq 0)$$

When is the acceleration 0?

$$v(t) = \frac{(1)(t^2 + 1) - t(2t)}{(t^2 + 1)^2} = \frac{t^2 + 1 - 2t^2}{(t^2 + 1)^2}$$

$$v(t) = \frac{1-t^2}{(t^2+1)^2}$$

$$a(t) = \frac{[-2t](t^2+1)^2 - (1-t^2)(2)(t^2+1)(2t)}{(t^2+1)^4}$$

$$a(t) = \frac{-2t(t^2+1)[t^2+1] + \overbrace{2(1-t^2)}^{2-2t}}{(t^2+1)^4}$$

$$= \frac{-2t(t^2+1)[3-t^2]}{\underbrace{[(t^2+1)^4]}_{\text{Never } 0}} \quad \underline{\text{set } 0}$$

$$\Rightarrow -2t = 0 \quad \text{OR} \quad t^2+1=0 \quad \text{OR} \quad 3-t^2=0$$

$$\boxed{t=0} \quad \text{Never} \quad t^2=3$$

$$t = \pm\sqrt{3}$$

$$\underline{\underline{t \geq 0}} \rightarrow \boxed{t = \sqrt{3}}$$

Higher Order Derivatives

As long as the derivative is diff, there is nothing stopping us from taking more $(\frac{d}{dx})$ s

$$3^{\text{rd}}: \quad \frac{d}{dx} \left(\frac{d^2 f}{dx^2} \right) = \frac{d^3 f}{dx^3}$$

$$[f''(x)]' = f'''(x)$$

$$4^{\text{th}}: \frac{d}{dx} \left(\frac{d^3 f}{dx^3} \right) = \frac{d^4 f}{dx^4}$$

$$[f'''(x)]' = f''''(x)$$

$$n^{\text{th}}: \frac{d}{dx} \left(\frac{d^{n-1} f}{dx^{n-1}} \right) \stackrel{\text{def}}{=} \frac{d^n f}{dx^n}$$

$$[f^{(n-1)}(x)]' \stackrel{\text{def}}{=} f^{(n)}(x)$$

NOTE: In a physics context, the derivative of acceleration $a(t)$ is called the jerk at t .
 $j(t) \stackrel{\text{def}}{=} a'(t) = v''(t) = s'''(t)$

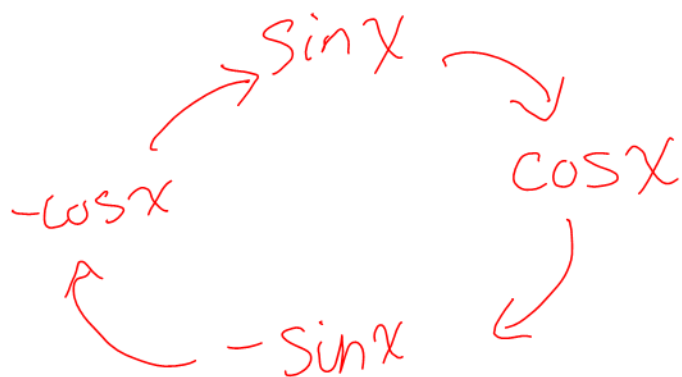
Example 4
 If $f^{(9)}(x) = 6 \sec(2x-5)$, find $f^{(10)}(x)$.

$$\begin{aligned} f^{(10)}(x) &= [f^{(9)}(x)]' = [6 \sec(2x-5)]' \\ &= 6 \sec(2x-5) \tan(2x-5) (2) \\ &= 12 \sec(2x-5) \tan(2x-5) \end{aligned}$$

Example 5 Examine the various derivative of $f(x) = \sin x$

$$f(x) = \sin x$$

$$\begin{aligned}
 f'(x) &= \cos x \\
 f''(x) &= -\sin x \\
 f'''(x) &= -\cos x \\
 f^{(4)}(x) &= \sin x
 \end{aligned}$$



Example 6 Repeat for $f(x) = x^3 + x^2 + x + 1$

$$f(x) = x^3 + x^2 + x + 1$$

$$f'(x) = 3x^2 + 2x + 1$$

$$f''(x) = 6x + 2$$

$$f'''(x) = 6$$

$$f^{(4)}(x) = 0$$

Example 7 Repeat for $f(x) = e^{2x}$

$$f(x) = e^{2x}$$

$$f'(x) = e^{2x} \cdot 2 \longrightarrow 2e^{2x}$$

$$f''(x) = 2e^{2x} \cdot 2 \longrightarrow 2^2 e^{2x}$$

$$f'''(x) = 2^2 e^{2x} \cdot 2 \longrightarrow 2^3 e^{2x}$$

$\vdots$

$$f^{(n)}(x) \longrightarrow 2^n e^{2x}$$

Ex 8 (HW 12, # 2) Find $g''(x)$ if

$$g(x) = 4e^{3x} \cos(4x)$$

$$g'(x) = 4e^{3x}(3) \cos(4x) + 4e^{3x}(-\sin(4x))(4)$$
$$= 12e^{3x} \cos(4x) - 16e^{3x} \sin(4x)$$

$$= \underline{4e^{3x}} [3 \cos(4x) - 4 \sin(4x)]$$

$$g''(x) = 12e^{3x} [3 \cos(4x) - 4 \sin(4x)]$$

$$+ 4e^{3x} [-12 \sin(4x) - 16 \cos(4x)]$$

$$= (36 - 64)e^{3x} \cos(4x) + (-48 - 48)e^{3x} \sin(4x)$$

$$= -28e^{3x} \cos(4x) - 96e^{3x} \sin(4x)$$